

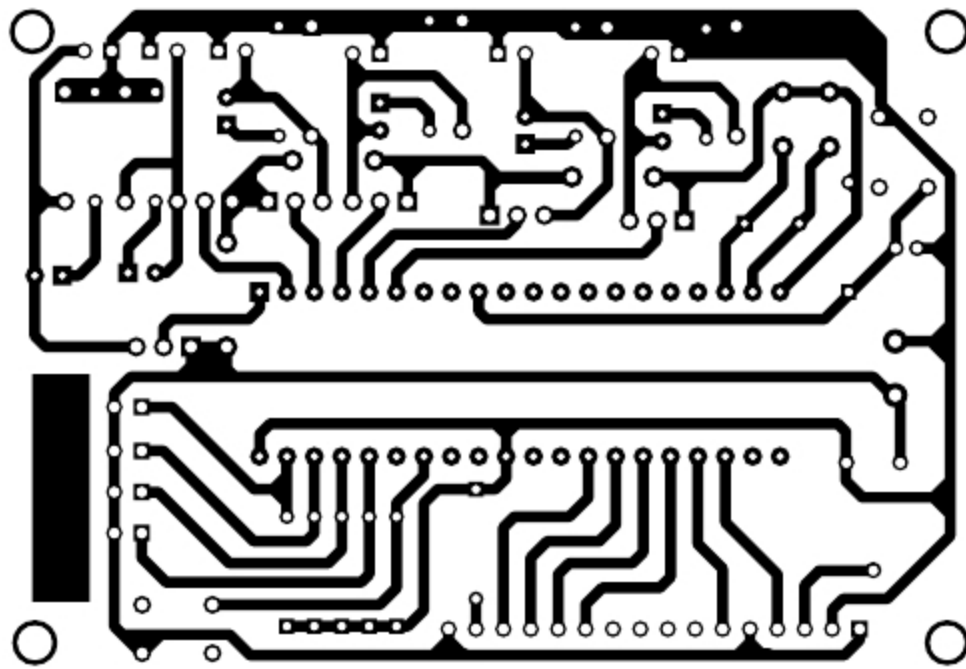


Fig. 8: Actual-size PCB layout for thin-film deposition technique using MCU

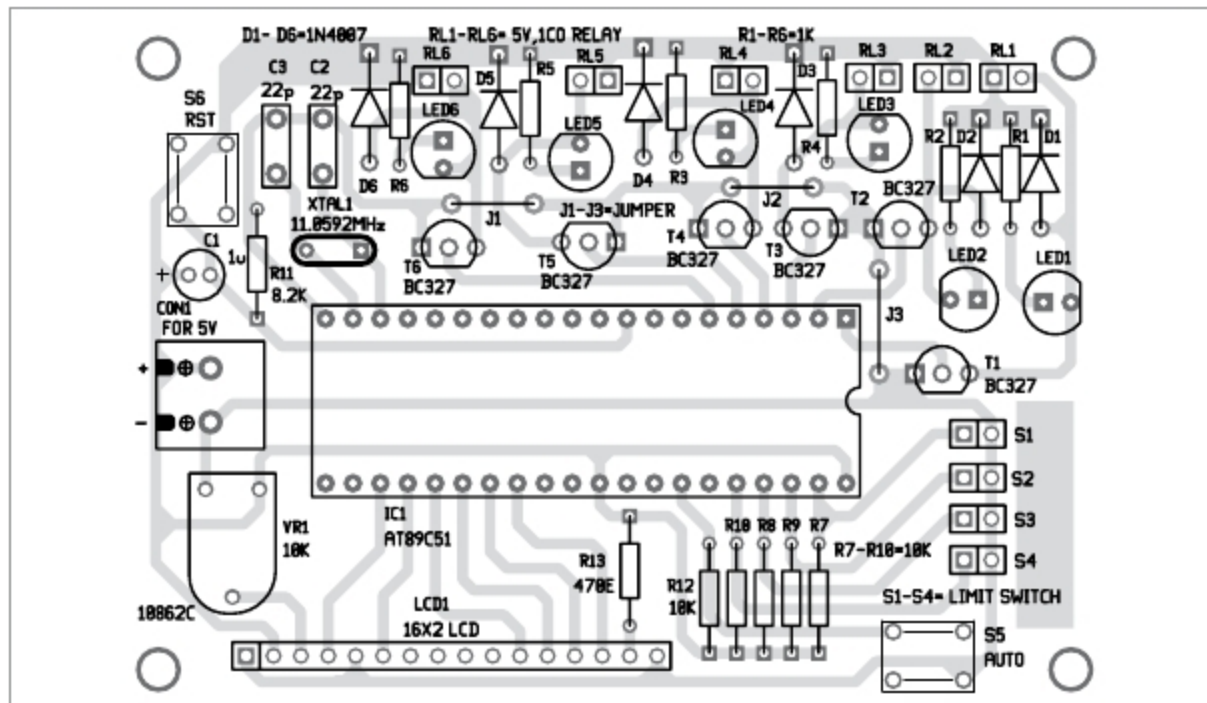


Fig. 9: Components layout for the PCB

or right directions (x direction) by pressing S4 to get appropriate delay of, say, one second, two seconds, three seconds and so on, followed by pressing S3.

3. Press S5 to start in auto mode. M1 starts rotating in CCW (RL1 is on and other relays are off) and the head moves towards S2.

4. When head touches S2, M1 stops for some time as per delay settings in step 2. And, M2 gets activated and the head moves downwards.

5. M1 gets activated in CW direction automatically. And, the head starts moving from left to right.

6. The cycle (steps 4 through 6) repeats until S3 or S4 is pressed.

7. Conditions for S3 and S4 are as follows. During run time, when S3 is pressed, the head moves in CW direc-

tion automatically. If S4 is pressed, the head immediately moves from the bottom towards home position.

**EFY note.** 1. The project was tested with 5V, low current rating relays. It is recommended to use proper relays with sufficient current and voltage ratings.

2. The source code for this project is available for free download at [source.efymag.com](http://source.efymag.com) **EFY**

Jitendra Jangid (left) is an electronics engineer. His hobbies include circuit designing and PCB making, and programming embedded systems like Arduino and 8051.



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